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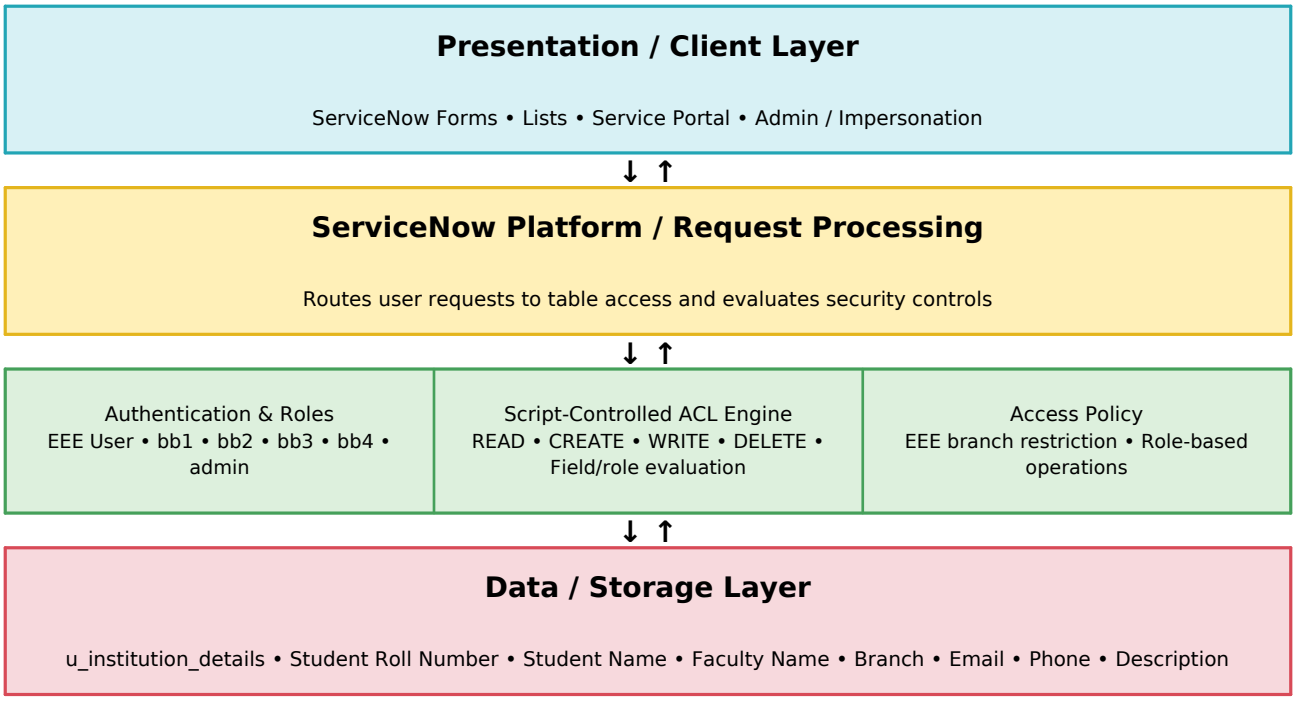
SKILL WALLET

Solution Architecture

Date	15 March 2026
Team ID	SWTID-2026-9550
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	5 Marks

Solution Architecture Diagram:

High-level architecture for the ServiceNow implementation. The design maps the client interface, ServiceNow request/security processing, ACL and role logic, and the Institution Details data layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation / Client Layer	Provides access to Student Records through ServiceNow forms, lists and Service Portal interfaces; supports admin impersonation for verification.	ServiceNow UI, Forms, Lists, Service Portal
Authentication & Roles	Controls which users can perform operations using custom roles bb1, bb2, bb3 and bb4, with admin receiving full access.	ServiceNow Users & Roles, Security Admin
Script-Controlled ACL Engine	Evaluates the ACL operation and user role. The READ script permits admin access, permits bb1 users, and denies other users.	ServiceNow ACL, Advanced Script, GlideSystem (gs)
Access Policy	Applies the EEE branch restriction for record visibility and progressively enables Create, Write and Delete through bb2, bb3 and bb4.	Record ACLs, Data Condition, Roles
Database / Table	Stores Institution Details records and the fields used by the access-control design, including Branch.	ServiceNow custom table: u_institution_details

Script-Controlled ACL – Access Flow

ACL Operation Matrix

Operation	Required Role	Access Behaviour
Read	bb1	EEE branch records are visible to users with bb1. Users without the role are denied; admin can view all records.
Create	bb2	Create ACL requires bb2. The project verification states users with bb1 + bb2 can work with EEE branch records.
Write	bb3	Write ACL requires bb3. Users with bb1 + bb2 + bb3 can view and edit EEE branch records.
Delete	bb4	Delete ACL requires bb4. Users with bb1 + bb2 + bb3 + bb4 can see, create, edit and delete EEE branch records.

Access Decision Flow

1. User Request User opens u_institution_details records.	→
2. ACL Evaluation ServiceNow checks the requested operation and the applicable ACL.	→
3. Role / Script Check Roles bb1–bb4 and the ACL script determine whether access is allowed.	→
4. Record Access Allowed: requested records/operation is available. Denied: access is blocked.	

Project Outcome

The implementation demonstrates how script-controlled READ, WRITE, CREATE and DELETE ACLs enforce record-level security in ServiceNow. User roles and record field values are evaluated together to control access and protect Institution Details records from unauthorized viewing, modification, creation and deletion.

Source Project Details

Team ID: SWTID-2026-9550 | Team Size: 5 | Platform: ServiceNow | Custom Table: u_institution_details